MSA680 - Lab 1

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1 Exploratory Analysis

In some medical setting, participants are given one of four different diets as treatments, and their cholesterol levels are measured before and after the treatments to determine whether the diet had an effect on their cholesterol levels. Eight participants are within each diet group, therefore there are a total of 32 data entries for pre-treatment cholesterol (PreChol) and post-treatment cholesterol (PostChol). The summary statistics of the data can be seen in Table 1 and Figure 1 below:

Diet	n	PreChol mean	PostChol mean	PreChol std.	PostChol std.	Chol mean Δ
1	8	240	194	53.1	16.9	-46.4
2	8	231	206	43.2	16.3	-24.6
3	8	225	218	40.0	17.0	-6.75
4	8	222	225	56.9	18.2	3.12

Table 1: Summary statistics for cholesterol levels

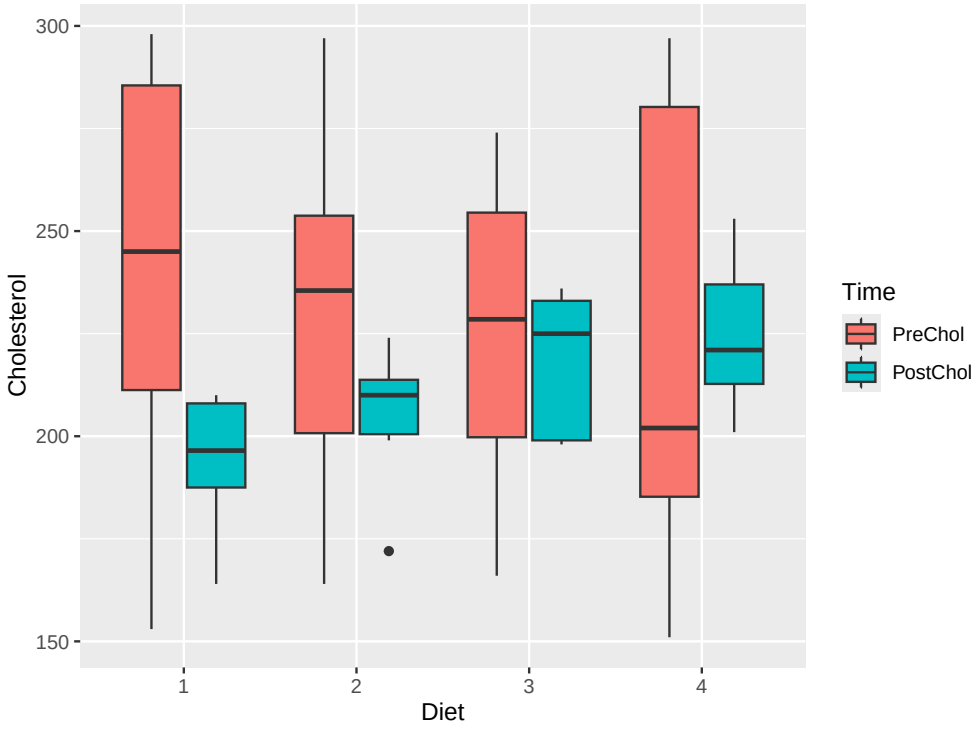


Figure 1: Boxplot of pre-treatment and post-treatment cholesterol levels by diet

As an overview, with the exception of diet 4, the pre-treatment levels of cholesterol (ranging from $220 \sim 240$) are higher than post-treatment (ranging from $190 \sim 225$). Diets 1 and 2 were associated with relatively large decreases in mean cholesterol at 46.4 units and 24.6 units respectively. Across the board, the spread in cholesterol levels across the subjects within a treatment group is larger pre-treatment (standard deviation between $40 \sim 50$) vs. post-treatment (standard deviation between $15 \sim 20$).

One hypothesis that could be tested therefore could be as follows:

$$\begin{cases} H_0: & \text{Diets are not significantly different in terms of post-diet cholesterol} \\ H_a: & \text{Some diets are significantly different in terms of post-diet cholesterol} \end{cases}$$

2 ANOVA Analysis

Considering only the post-diet data, we can perform a one-way ANOVA analysis that determines whether the four diet types D_1, D_2, D_3, D_4 give the same mean post-treatment cholesterol ($H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$) or not ($H_a : \mu_i \neq \mu_j, 1 \leq i \neq j \leq 4$). Here, we are performing an F test at the $\alpha = 0.05$ level significance for the test statistic $F(0.05, 4-1, 32-4)$. The results are summarized in Table 2 below.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	Sigificance
Diet	3	4594	1531.3	5.235	0.00538	**
Residuals	28	8190	292.5			

Table 2: Results from ANOVA analysis

Since $0.00538 < 0.05$, we reject the null hypothesis that all four diets give the same mean post-treatment cholesterol levels at the $\alpha = 0.05$ significance level.

We can use the lm function to fit a linear model to the same data as above. The response is the post-diet cholesterol, while the covariate is the categorical variable of diet type. The results are summarized in Table 3 below. Feeding the output of the lm function to anova() in R gives the same output above (see attached code for reference).

Parameter	Estimate	Std. Error	t value	$\Pr(>\|t\|)$	Significance
(Intercept)	193.625	6.047	32.022	$< 2 \times 10^{-16}$	***
Diet2	12.500	8.551	1.462	0.15493	
Diet3	24.875	8.551	2.909	0.00703	**
Diet4	31.250	8.551	3.654	0.00105	**

Table 3: Results from the lm function

The baseline parameter represents the average post-diet cholesterol levels (193.625 units) for Diet 1. This value is significantly different from 0, which is expected as the cholesterol for any person is likely non-zero. The other parameters represent the difference in post-diet cholesterol levels compared with the baseline diet 1. For example, the average post-diet cholesterol levels for diet 4 is expected to be 31.250 units higher than diet 1. Note that at the $\alpha = 0.05$ level, only diet 3 and diet 4 post-diet cholesterol levels are significantly different from diet 1.

Using the emmeans() function, we can find the 95% confidence intervals for the parameter estimates of the mean post-diet cholesterol levels for the four diets. They are summarized in Table 4 below.

Diet	lsmean	95% CI
Diet 1	194	[181, 206]
Diet 2	206	[194, 219]
Diet 3	218	[206, 231]
Diet 4	225	[212, 237]

Table 4: 95% Confidence Intervals for Post-diet cholesterol mean

Note that we can calculate the 95% confidence interval by hand using the output of the lm function. The 95% CI is defined as:

$$\left[\bar{Y} - t_{\alpha/2,n-4} \cdot \frac{s}{\sqrt{n}}, \bar{Y} + t_{\alpha/2,n-4} \cdot \frac{s}{\sqrt{n}}\right]$$

Where $\frac{s}{\sqrt{n}}$ is the standard error reported in table 3. $t_{(0.025,32-4)} = 2.048$, so the 95% CI for

Diet 1 can thus be calculated as follows:

$$[193.625 - 2.048 \cdot 6.047, 193.625 + 2.048 \cdot 6.047] = [181.2, 206.0]$$

Which matches with the 95% CI output of the emmeans() function.

3 ANCOVA Analysis, Equal Slopes

Now considering both pre-diet post-diet data, we now consider a 2-covariate model to check whether the four diet types D_1, D_2, D_3, D_4 (categorical covariate), in addition to the pre-diet cholesterol (continuous covariate) have a significant effect on post-diet cholesterol (response variable) in an ANCOVA analysis which involves both categorical groups and linear regression. First, we employ an equal-slopes model, which further assumes that the "rate" of cholesterol change, dependent on the pre-diet levels of cholesterol, is equal among all four diets. In other words, $Y_{ij} = \alpha_i + \beta x_{ij} + \varepsilon_{ij}$, where α_i represents the intercepts for each diet, β represents the post-diet cholesterol change expected per pre-diet cholesterol unit. The results are summarized in Table 5 below.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	Significance
PreChol	1	130	130.4	0.430	0.51752	
Diet	3	4465	1488.4	4.908	0.00753	**
Residuals	27	8188	303.2			

Table 5: Results from ANCOVA analysis (equal slopes model)

Looking at the results at the 95% confidence level, we can say that under the equal slope assumption with the above two covariates inside the model, diet type significantly "influences" post-diet cholesterol levels, but pre-diet cholesterol levels does not. A better interpretation will follow looking at the outputs of the linear model.

Again, we can use the lm function to fit a linear model to the same data as above. The response is the post-diet cholesterol, while the covariates are the categorical variable diet

type, and the continuous variable pre-diet cholesterol. The baseline chosen was the diet 1 post-diet cholesterol level. The results are summarized in Table 6 below. Feeding the output of the lm function to anova() in R, again, gives the same output as above (see attached code for reference).

Parameter	Estimate	Std. Error	t value	$\Pr(>\|t\|)$	Significance
(Intercept) (α_{D1})	194.908374	17.328472	11.248	1.08×10^{-11}	***
PreChol (β)	-0.005347	0.067491	-0.079	0.93743	
Diet2 ($\alpha_{D2} - \alpha_{D1}$)	12.450537	8.729374	1.426	0.16525	
Diet3 ($\alpha_{D3} - \alpha_{D1}$)	24.796126	8.763746	2.829	0.00869	**
Diet4 ($\alpha_{D4} - \alpha_{D1}$)	31.152410	8.793711	3.543	0.00146	**

Table 6: Results from the lm function (equal slopes model)

The intercept states that for a participant with 0 pre-diet cholesterol level who underwent diet 1, the estimated post-diet cholesterol level is 194.9. The parameter Diet2 = 12.45 states that for a participant with 0 pre-diet cholesterol level who underwent diet 2, the estimated post-diet cholesterol level is $\alpha_{D2} = 194.9 + 12.45 = 207.4$. It is the same for the parameters Diet3 and Diet4. It seems like the Diet2 is insignificantly different from baseline Diet1, while Diet3 and Diet4 have both significantly higher estimated post-diet cholesterol levels than baseline Diet1. The β parameter indicates that for each unit of pre-diet cholesterol, the post-diet cholesterol is expected to decrease by -0.005347 units. In other words, for a participant with 200 pre-diet cholesterol who underwent diet 1, their post-diet cholesterol would be estimated to be $y = 194.9 - 0.005347 \times 200 \approx 193.8$. However, β is an insignificant effect due to the high p value, meaning that pre-diet cholesterol levels are not a significant predictor of post-diet cholesterol levels. This can be further visualized in Figure 2 below.

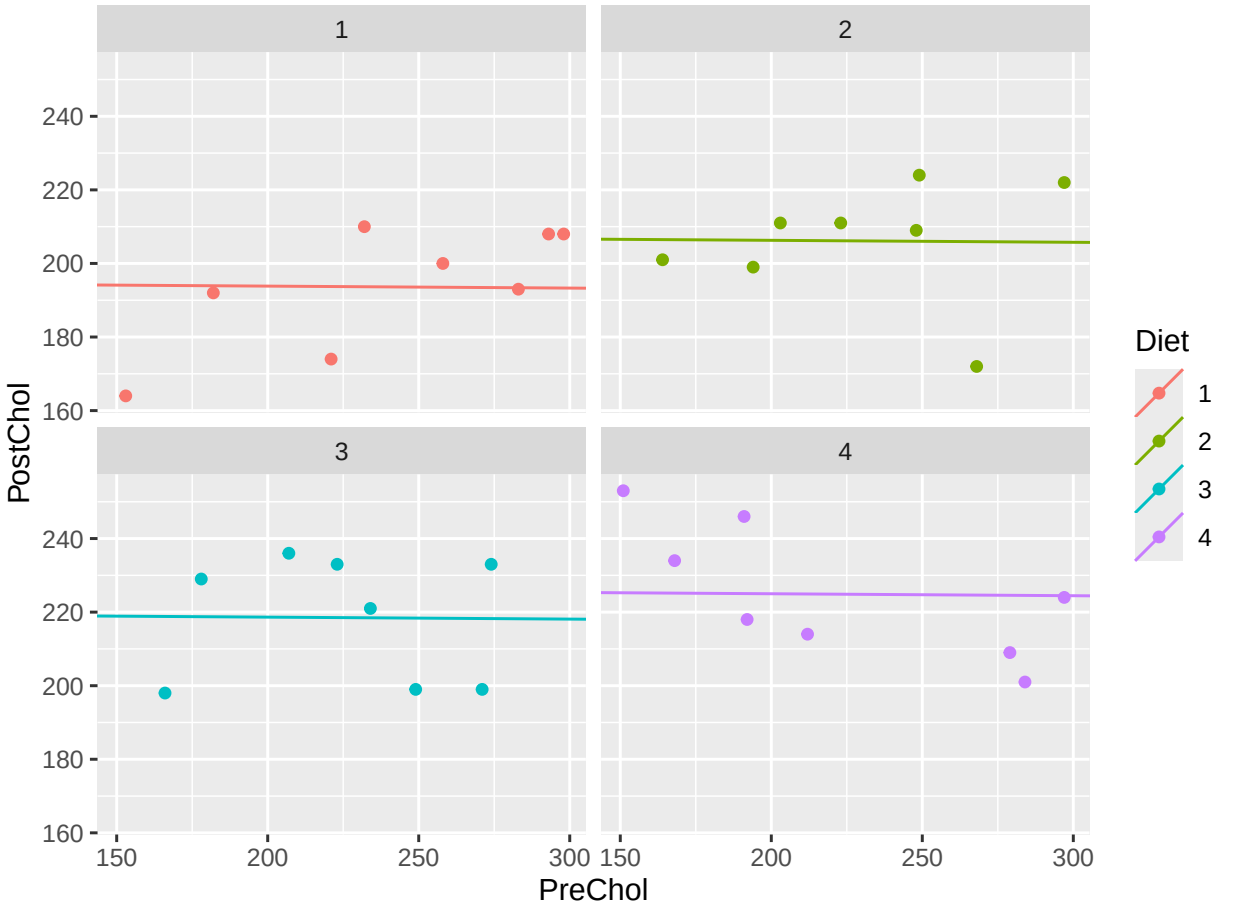


Figure 2: Scatterplot of Post-diet Cholesterol vs. Pre-diet Cholesterol (equal slopes model)

From Figure 2 we can see that the intercepts seem to be very different depending on diet, but the gradient itself seems very flat. This again corresponds with the interpretation that in this equal slopes model, diet type is a significant predictor of post-diet cholesterol, while pre-diet cholesterol is not. For this model, the R^2 was 0.3595, suggesting a moderate correlation.

Using the emmeans() function, we can obtain the post-diet cholesterol lsmeans, corrected for pre-diet cholesterol levels as seen in Table 7. These results are different from the results obtained from the ANOVA analysis with just post-diet cholesterol, as the lsmeans are corrected for different levels of pre-diet cholesterol which are now in the model.

Diet	lsmean	95% CI
Diet 1	193.6815	[180.96, 206.40]
Diet 2	206.1320	[193.50, 218.77]
Diet 3	218.4476	[205.83, 231.12]
Diet 4	224.8339	[212.16, 237.51]

Table 7: 95% Confidence Intervals for Post-diet cholesterol lsmeans (corrected for Pre-diet levels)

4 ANCOVA Analysis, Unequal Slopes

For the next model, we will consider interactions between diet group and pre-diet cholesterol added to the previous model. This means that the "rate" of cholesterol change from pre-diet to post-diet is no longer assumed to be the same among the four diet types, and graphically the slopes will be unequal. In other words, $Y_{ij} = \alpha_i + \beta_i x_{ij} + \varepsilon_{ij}$, where α_i represents the intercepts for each diet, β_i represents the post-diet cholesterol change expected per pre-diet cholesterol unit for each diet. The results are summarized in Table 8 below.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	Significance
PreChol	1	130	130.4	0.535	0.47170	
Diet	3	4465	1488.4	6.103	0.00308	**
PreChol:Diet	3	2335	778.3	3.191	0.04173	*
Residuals	24	5853	243.9			

Table 8: Results from ANCOVA analysis (unequal slopes model)

Looking at the results at the 95% confidence level, both diet type and the interaction effect of diet type and pre-diet cholesterol levels are significant. Therefore, while pre-diet cholesterol itself was not significant at the 95% confidence level, due to the significant effect of the interaction term, we can say that both diet-type and pre-diet cholesterol levels have an effect on post-diet cholesterol levels.

Again, we can use the lm function to fit a linear model. The baseline for intercept was the diet 1 cholesterol level, and the baseline for slope was the diet 1 cholesterol level as well. The results are summarized in Table 9 below, and the graphs with the parameter estimates are shown in Figure 3 below.

Parameter	Estimate	Std. Error	t value	$\Pr(>\|t\|)$	Significance
(Intercept) (α_{D1})	137.6323	27.2651	5.048	3.68×10^{-5}	***
PreChol (β_{D1})	0.2333	0.1113	2.097	0.046701	*
Diet2 ($\alpha_{D2} - \alpha_{D1}$)	58.1038	42.0641	1.381	0.179908	
Diet3 ($\alpha_{D3} - \alpha_{D1}$)	86.1007	43.3530	1.986	0.058565	.
Diet4 ($\alpha_{D4} - \alpha_{D1}$)	138.9710	36.1061	3.849	0.000771	***
PreChol:Diet2 ($\beta_{D2} - \beta_{D1}$)	-0.1883	0.1763	-1.068	0.296104	
PreChol:Diet3 ($\beta_{D3} - \beta_{D1}$)	-0.2565	0.1848	-1.388	0.177937	
PreChol:Diet4 ($\beta_{D4} - \beta_{D1}$)	-0.4666	0.1522	-3.066	0.005296	**

Table 9: Results from the lm function (unequal slopes model)

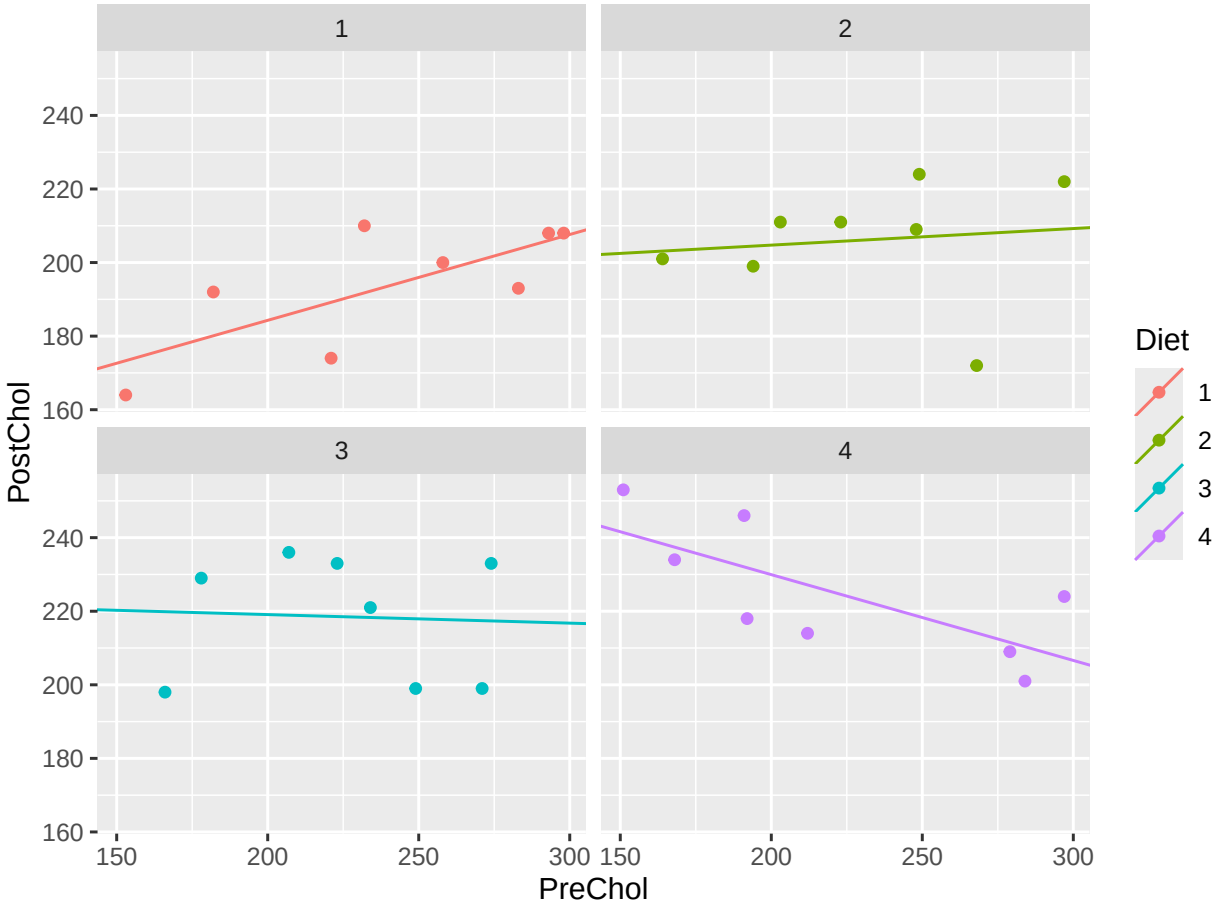


Figure 3: Scatterplot of Post-diet Cholesterol vs. Pre-diet Cholesterol (unequal slopes model)

Looking at Figure 3 (without looking at confidence intervals and significance values), it seems that within the pre-diet cholesterol range of 150-300 units, it seems like diets 1 and 2 gave lower post-diet cholesterol than diets 3 and 4; on average, the post-diet cholesterol

was lower than 220 for diets 1 and 2, while at or higher tthan 220 for diets 3 and 4.. Diets 2 and 3 had relatively "flat" slopes, meaning that the post-diet cholesterol level is independent of pre-diet cholesterol level. Diets 1 and 4 had non-zero slopes, meaning that the post-diet cholesterol levels depend on the pre-diet cholesterol levels.

Looking more closely at the parameter estimates in Table 9, firstly, we can see that PreChol (β_{D1}) slope is significantly different from 0 at $\beta_{D1} = 0.2333$. This means that for someone who underwent diet 1, for each unit increase in pre-diet cholesterol, the post-diet cholesterol will be expected to increase by 0.2333 units. Looking at the interaction terms, it seems like the slopes for diet 2 and diet 3, while lower than the slope of the diet 1 baseline, are not significantly different. Graphically we can see the slopes for diet 2 and 3 are slightly "flatter" than diet 1, and it seems like both slopes are close to 0 (e.g. $\beta_{D2} = 0.2333 - 0.1883 = 0.05$) and therefore for these two diets the post-diet cholesterol may be independent amount of pre-diet cholesterol. For diet 4, the slope is significantly different from diet 1 and is significantly more negative than diet 1 ($p = 0.005296$), and for each unit increase in pre-diet cholesterol, the post-diet cholesterol is expected to decrease by $\beta_{D4} = 0.2333 - 0.4666 = -0.2333$ units.

Looking at the baseline intercept α_{D1}, we can see that for someone with pre-diet 1 cholesterol of 0, the post-diet cholesterol is expected to be 137.6323. Compared to this, the diet 2 and 3 intercepts are not significantly different from that of diet 1, however the diet 4 cholesterol is significantly higher than baseline at $\alpha_{D4} = 137.6323 + 138.9710 = 276.6033$. However this is not really useful in terms of interpretation, as it is extremely unrealistic for someone to have a 0 cholesterol level.

Using the emmeans() function, again we can obtain the post-diet cholesterol lsmeans, corrected for pre-diet cholesterol levels as seen in Table 10. These lsmeans for the post-diet cholesterol levels are corrected to the mean pre-diet cholesterol levels for each diet. These results are different from the results obtained from previous equal slopes model, as interactions are now present in the model. For this model, the R^2 was 0.5421, a stronger

correlation than the previous model.

Diet	lsmean	95% CI
Diet 1	191.1607	[179.51, 202.81]
Diet 2	206.0659	[194.66, 217.47]
Diet 3	218.4027	[206.94, 229.87]
Diet 4	223.0817	[221.57, 234.60]

Table 10: 95% Confidence Intervals for Post-diet cholesterol lsmeans (corrected for Pre-diet levFigure els)

If we were to fit the same unequal slopes model as above, but replace the pre-diet cholesterol covariate with the difference with mean pre-diet cholesterol (mPreChol = PreChol - mean(PreChol)), then the graph will look the same as Figure 3 above. This is because we are basically recentering the x-axis, so that the y-intercept will occur at the mean PreChol level across diets instead of at PreChol=0 in the previous model. The parameter estimates are included below.

Parameter	Estimate	Std. Error	t value	$\Pr(>\|t\|)$	Significance
(Intercept)	191.1607	5.6449	33.864	$< 2 \times 10^{-16}$	***
mPreChol	0.2333	0.1113	2.097	0.046701	*
Diet2	14.9052	7.8982	1.887	0.071290	.
Diet3	27.2420	7.9203	3.440	0.002139	**
Diet4	31.9210	7.9363	4.022	0.000498	***
mPreChol:Diet2	-0.1883	0.1763	-1.068	0.296104	
mPreChol:Diet3	-0.2565	0.1848	-1.388	0.177937	
mPreChol:Diet4	-0.4666	0.1522	-3.066	0.005296	**

Table 11: Results from the lm function (unequal slopes model centered at mPreChol)

Again, here we can see that the slopes and interactions are exactly the same as previous results, and represent the effects of a change in one unit of pre-diet cholesterol on post-diet cholesterol. However, the intercepts are different but more interpretable. 191.1607 is the estimated post-diet cholesterol level for someone who underwent diet 1 and had a pre-diet cholesterol level at mPreChol=229.4375. Similarly, 14.9052+191.1607=206.0659 is the estimated post-diet cholesterol level for someone who underwent diet 2 and had a pre-diet cholesterol level at mPreChol=229.4375

If we take a deeper look, we can calculate lsmeans and confidence intervals for post-diet cholesterol levels, given that a participant has a pre-diet cholesterol level at 190 units (below desired) or 250 units (high-risk). The results are summarized in Table 12.

Diet	PreChol=190 lsmean	95% CI	PreChol=250 lsmean	95% CI
Diet 1	181.96	[165.78, 198.14]	195.96	[184.33, 207.58]
Diet 2	204.29	[188.10, 220.48]	206.99	[194.37, 219.62]
Diet 3	219.3189	[203.66, 234.98]	217.93	[204.26, 231.59]
Diet 4	232.28	[219.01, 245.55]	218.29	[205.38, 231.19]

Table 12: lsmeans and 95% CI for Post-diet cholesterol lsmeans for different risk-groups

Since 200 is the desired level of cholesterol, it seems like diet 2 is the best choice that will lead to a good cholesterol level for both groups with low pre-diet cholesterol levels, as well as groups with high pre-diet cholesterol levels.

5 SAS ANCOVA Analysis, Unequal Slopes

Fitting the same unequal slopes model in the previous section, we can get the same results in SAS (as seen in Figures 4 and 5 below) compared with in R (as seen in Figure 3 and Table 10 above).

Diet	PostChol LSMEAN	95% Confidence Limits	
1	191.160738	179.510264	202.811212
2	206.065908	194.664643	217.467173
3	218.402716	206.936274	229.869159
4	223.081714	211.568077	234.595351

Figure 4: Scatterplot of Post-diet Cholesterol vs. Pre-diet Cholesterol (equal slopes model)

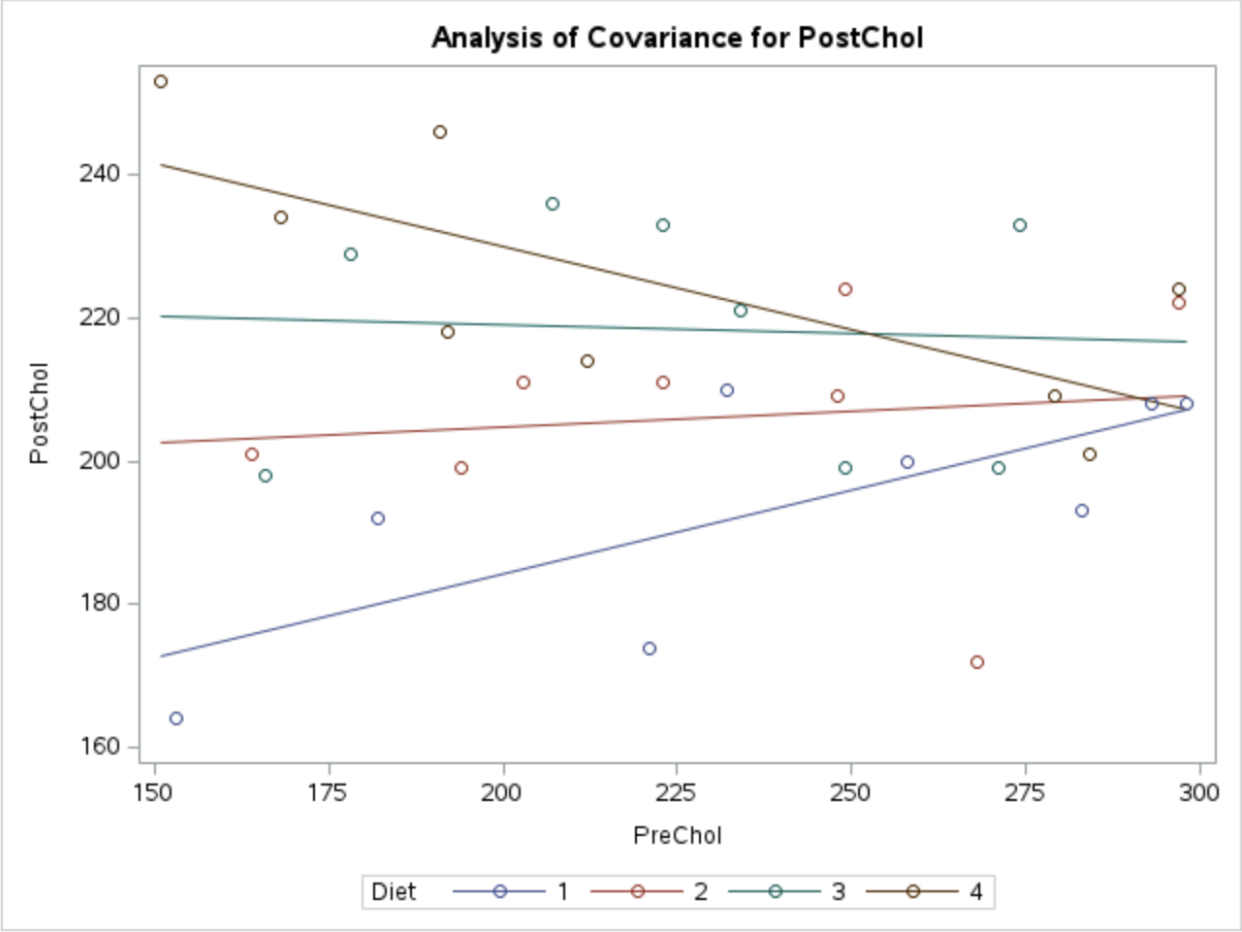


Figure 5: Scatterplot of Post-diet Cholesterol vs. Pre-diet Cholesterol (equal slopes model)

Furthermore, at pre-diet cholesterol levels of 190 and 250, the lsmeans and 95% confidence intervals given by SAS (Figures 6 and 7 below) are also the same as the results given by R (Table 12 above).

Diet	PostChol LSMEAN	95% Confidence Limits	
1	181.959856	165.784134	198.135578
2	204.290336	188.100755	220.479917
3	219.318925	203.660434	234.977416
4	232.281416	219.010592	245.552240

Figure 6: PreChol=190 lsmeans

Diet	PostChol LSMEAN	95% Confidence Limits	
1	195.958029	184.333755	207.582303
2	206.991682	194.367733	219.615630
3	217.925010	204.260798	231.589223
4	218.285039	205.382434	231.187644

Figure 7: PreChol=250 lsmeans

Finally, we can look at pairwise comparisons of post-diet cholesterol levels for pre-diet cholesterol levels at 190 and 250 - this will allow us to directly compare whether two diets will

give significantly different post-diet cholesterol when the pre-diet levels are held constant. The results are summarized in Figures 8 and 9 below.

Least Squares Means for Effect Diet				
i	j	Difference Between Means	95% Confidence Limits for LSMean(i)-LSMean(j)	
1	2	-22.330480	-45.216208	0.555247
1	3	-37.359069	-59.872228	-14.845910
1	4	-50.321560	-71.244485	-29.398636
2	3	-15.028589	-37.551707	7.494530
2	4	-27.991080	-48.924721	-7.057440
3	4	-12.962491	-33.488161	7.563178

Figure 8: PreChol=190 pdiff

Least Squares Means for Effect Diet				
i	j	Difference Between Means	95% Confidence Limits for LSMean(i)-LSMean(j)	
1	2	-11.033653	-28.194300	6.126995
1	3	-21.966981	-39.906726	-4.027236
1	4	-22.327010	-39.693673	-4.960348
2	3	-10.933328	-29.536414	7.669757
2	4	-11.293357	-29.344432	6.757717
3	4	-0.360029	-19.153324	18.433266

Figure 9: PreChol=250 pdiff

To look at which pairs give significantly post-cholesterol levels, the same results from the above two figures can be interpreted more easily in a graphical format below (Figures 10 and 11).

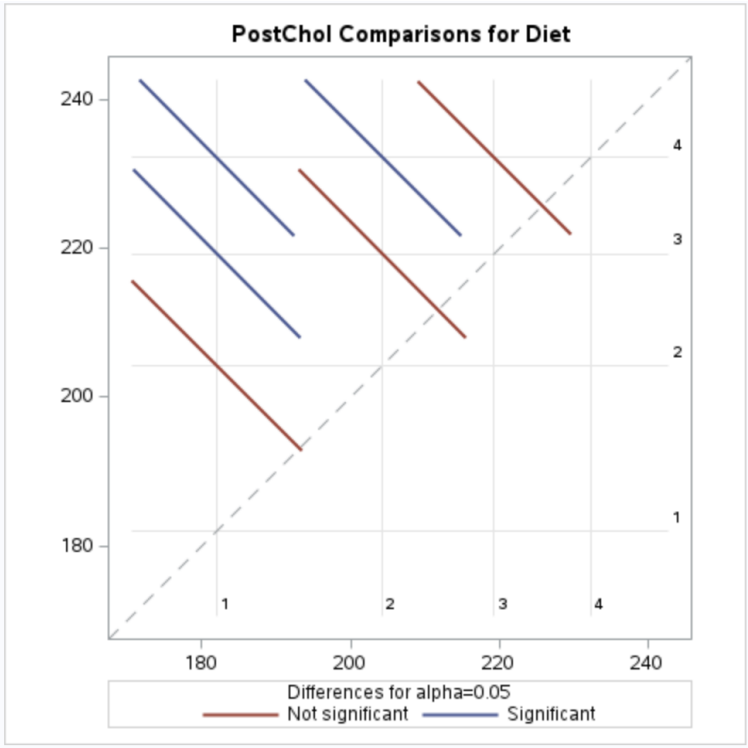


Figure 10: PreChol=190 pdiff graph (95% significance)

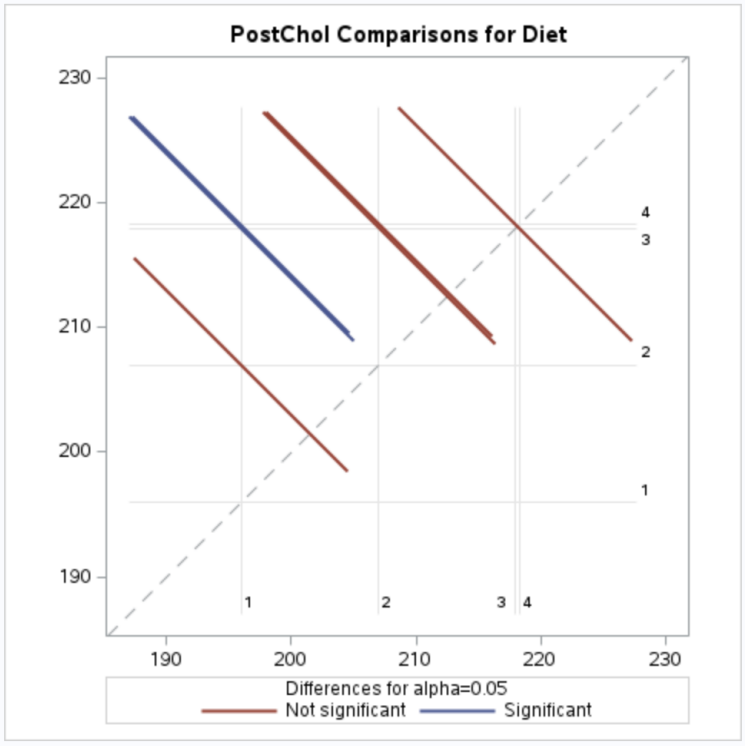


Figure 11: PreChol=250 pdiff graph (95% significance)

When pre-diet cholesterol is 190 (below optimal), the following diet pairs are expected to give significantly different post-diet cholesterol levels with 95% confidence: diets 1, 3; diets 1, 4; diets 2, 4. Similarly, when pre-diet cholesterol is 250 (above optimal), the following diet pairs are expected to give significantly different post-diet cholesterol levels with 95% confidence: diets 1, 3; diets 1, 4.